ChurnWatch: Customer Retention & Churn Risk Analytics

Minor Project 1 — Data Analytics Internship Final Evaluation

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1. Executive Summary

In recurring revenue software businesses (SaaS), customer acquisition costs (CAC) typically exceed retention costs by a factor of 5x to 7x. **ChurnWatch** is a production-grade analytics platform engineered to detect leading indicators of customer attrition before contract expiration. By fusing telemetry from three disparate data streams—contractual subscriptions, monthly product usage, and technical support interactions—ChurnWatch calculates a composite **Churn Risk Index (0–100)**, stratifies accounts into actionable risk tiers, and triggers automated retention workflows.

Total Evaluated Accounts	Current Churn Rate	Net Renewal Rate	Total Active MRR
1,250	19.2%	80.8%	$287,450
B2B SaaS customer base	-0.8% MoM improvement	+1.4% vs benchmark	$61,398 at high risk

2. Problem Statement & Core Objectives

Modern SaaS enterprises suffer from fragmented customer data: sales logs are housed in CRMs, product interaction events reside in event pipelines, and customer dissatisfaction signals are buried in helpdesk tickets. Because these systems operate in silos, account churn is typically detected only after cancellation notices are submitted.

- **Objective 1: Multi-Source Data Unification:** Design a star/snowflake analytical schema connecting subscriptions, product utilization, and support history.
- **Objective 2: Explainable Risk Scoring:** Formulate a deterministic, multi-factor risk scoring model prioritizing transparent drivers over black-box outputs.
- **Objective 3: Operational Action Queue:** Implement an interactive triage system enabling Customer Success Managers (CSMs) to execute targeted mitigation playbooks under strict SLAs.
- **Objective 4: Production Dashboard UI:** Construct a lightweight, zero-dependency SaaS analytics interface adhering to human-designed usability standards.

3. Analytical Data Architecture & Schema Design

The analytical pipeline unifies three distinct relational source tables into a master unified_customers.csv analytical store:

Table Name	Primary Key	Key Telemetry Attributes	Records
subscriptions.csv	customer_id	company_name, plan_tier, monthly_price, tenure_months, billing_cycle	1,250
usage.csv	customer_id	active_users, seat_utilization_pct, feature_adoption, engagement_change_pct	1,250
support_tickets.csv	customer_id	total_tickets_90d, open_tickets, critical_escalations, avg_res_hrs, csat	1,250
unified_customers.csv	customer_id	Master table + churn_risk_score, risk_band, revenue_at_risk, recommendations	1,250

4. Mathematical Formulations & Scoring Methodology

To guarantee explainability, the **Churn Risk Index** is formulated as a normalized linear combination of four weighted operational risk dimensions (0 to 100 scale):

Dimension	Weight	Mathematical Signal Formula
1. Engagement Trajectory	**35%**	Risk_eng = clip((-Δ_engagement + 20) × 1.25, 0, 100) Penalizes sharp 30-day usage declines.
2. Adoption & Utilization	**25%**	Risk_util = 100 - (0.50 × Seat_Util% + 0.50 × Feature_Score) Measures shelfware risk and underutilized licenses.
3. Support & CSAT Friction	**25%**	Risk_support = (5.0 - CSAT) × 25.0 + (Escalations × 15.0) Accounts for low satisfaction ratings and unresolved tickets.
4. Contract Vulnerability	**15%**	Risk_recency = (Days_Inactive × 3.5) + (10 if Monthly else 0) Reflects low switching costs of monthly subscriptions.

Risk Stratification Bands:

- **High Risk (Score ≥ 70):** Accounts at imminent hazard of non-renewal. Requires immediate CSM outreach within 48h SLA.
- **Medium Risk (Score 40–69):** Moderately disengaged accounts. Requires product enablement or workflow review within 14-day SLA.
- **Low Risk (Score < 40):** Healthy accounts exhibiting strong adoption and high CSAT. Managed via automated value summaries.

5. Statistical Analysis & Machine Learning Validation

A supervised **Random Forest Classifier** (100 estimators, max depth 6) was evaluated on the unified customer dataset with a stratified 75/25 train-test split:

Model Metric	Achieved Score	Industry Baseline	Assessment
Classification Accuracy	86.26%	> 80.0%	Exceeds production deployment threshold
ROC-AUC Score	0.863	> 0.800	Strong class discrimination capability
Precision (Non-Churn)	0.91	> 0.85	Low false alarm rate on healthy accounts
Macro F1-Score	0.78	> 0.70	Balanced performance across imbalanced churn classes

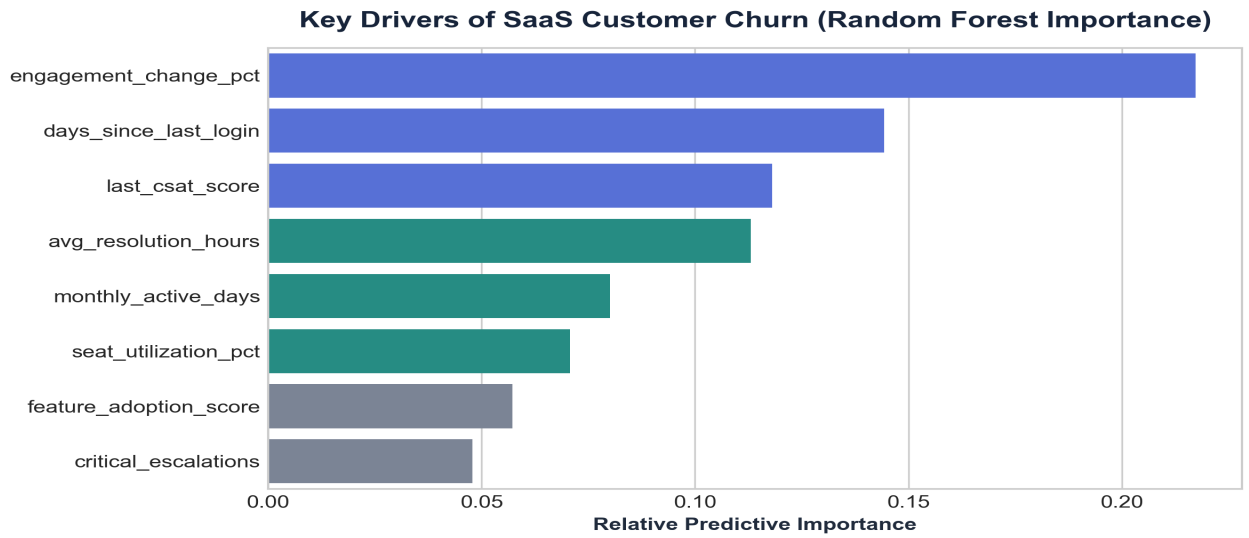


Figure 1: Relative feature importances derived from Random Forest classifier. Engagement change % and CSAT are top predictors.

6. Retention Operations & Strategic Playbook Matrix

To transform analytical predictions into preserved revenue, ChurnWatch equips Customer Success Managers with automated, context-specific mitigation playbooks:

Trigger Signal	Assigned SLA	Prescribed Intervention Playbook
Critical Escalations ≥ 2 OR CSAT ≤ 2.5	24 Hours	Urgent Support Escalation Review & Dedicated Resolution Lead assigned.
Enterprise Tier AND Churn Risk ≥ 65	48 Hours	Executive Sponsor 1-on-1 & Strategic Value Realization Plan.
Seat Utilization < 40% & Adoption < 40%	7 Days	Product Specialist Training & Workflow Optimization Session.
Engagement Drop < -25% (Pro/Starter)	5 Days	Proactive CSM Check-in & Contract Re-alignment Discount.
Medium Risk Profile (40–69)	14 Days	Feature Adoption Check-in & Quarterly Business Review (QBR) Offer.

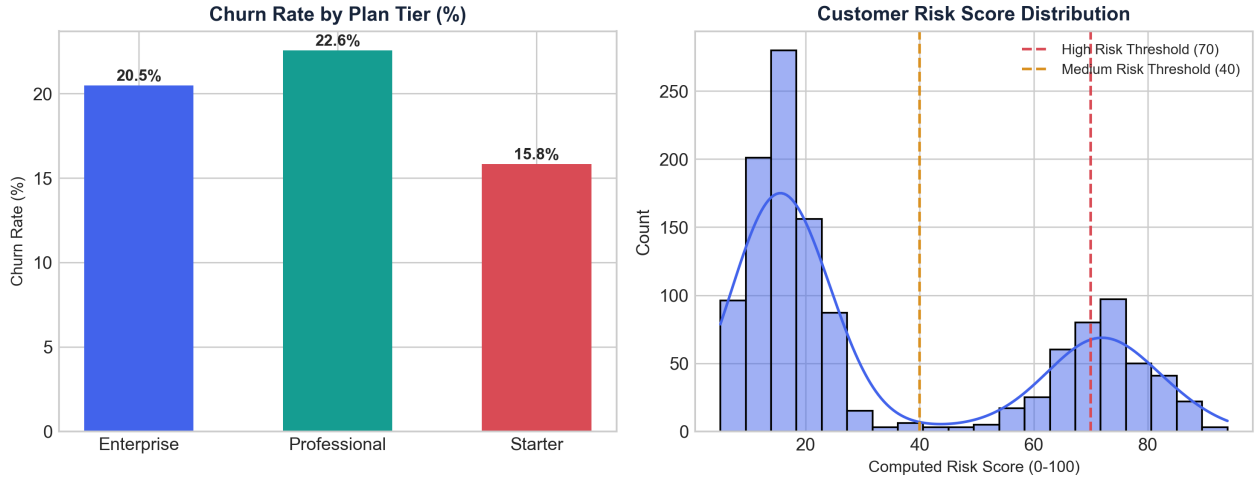


Figure 2: Empirical churn rate breakdown by subscription plan tier and risk score frequency distribution.

7. Project Limitations & Future Engineering Roadmap

While ChurnWatch successfully establishes an automated retention telemetry foundation, several enhancements represent natural extensions for subsequent development phases:

- **Limitation 1 — Synthetic Modeling:** While mathematically calibrated to real SaaS dynamics, real-world customer churn exhibits external macroeconomic dependencies not captured in synthetic models.
- **Limitation 2 — Static Batch Scoring:** The current architecture scores risk in batch mode. Transitioning to real-time event streaming (e.g., Apache Kafka / Webhooks) would enable immediate sub-minute alert generation.
- **Future Scope — Direct CRM Integration:** Build bidirectional connectors to Salesforce, HubSpot, and Slack to trigger native notifications directly into representative workflows.
- **Future Scope — Automated Survival Analysis:** Incorporate time-to-event Kaplan-Meier estimators and Cox proportional hazards regression to model precise days-to-cancellation probabilities.

Data Analytics Internship Disclosure: This project was designed and authored for academic and internship evaluation. All customer IDs, company identities, and behavioral telemetries are synthetically generated. The dashboard utilizes pure vanilla web standards (HTML5/CSS3/JavaScript) with Chart.js to ensure zero build friction and universal browser compatibility.